

Naser Kazemi

✉ naserkazemi2002@gmail.com | 🏠 naser-kazemi.github.io | 📍 Sharif University of Technology, Tehran, Iran

Education

Sharif University of Technology

Bachelor in Computer Engineering, Minor Degree in Applied Mathematical Science

Overall GPA: 19.27/20, Major GPA: 19.72/20, [Transcript](#)

Tehran, Iran

Sep. 2020 - Ongoing

Allameh Dehkhoda High School

Diploma in Mathematics and Physics

Overall GPA: 19.78/20

Noorabad, Iran

Sep. 2018 - 2020

Research Interests

- Machine Learning
- Reinforcement Learning
- Large Language Models
- Foundation and World Models
- Computer Vision
- Optimization

Research Experience

Interactive Video Generation and World Models

INSATIT, Under the Supervision of Prof. Luc Van Gool & Prof. Danda Pani Paudel

Sofia, Bulgaria

July. 2024 - ongoing

Advancing world models by integrating agent exploration, introducing unseen yet meaningful data that enables learning on new samples:

- We contribute a fully functional, open-source implementation of the Genie model. Our enhanced model, GenieRedux-G-TA, leverages reinforcement learning agents for data generation, reducing the need for random or human-collected data and improving adaptability across scenarios. Tested on the Coinrun case study, GenieRedux-G-TA shows superior visual fidelity, control, and scalability to new environments. (NeurIPS Workshop on D3S3)
- Proposed a novel framework for training multi-environment world models using uncertainty-driven exploration agents, achieving improved video fidelity and controllability without relying on costly action demonstrations or environment-specific rewards. (Submitted to CVPR 2025)

Multi-Objective Optimization Evolutionary Algorithms

Max Planck Institutes for Informatics, Under the Supervision of Prof. Vahid Babaie

Saarbrücken, Germany

July. 2023 - Sep. 2023

This project aimed to enhance solution diversity in multi-objective optimization by applying a Monte Carlo method across evolutionary algorithms, including NSGA-II, NSGA-III, and other typical algorithms. The method involved random segmentation of the objective space and customized sorting, which showed improved solution spread and diversity, especially in complex problem spaces.

Publications

Learning Generative Interactive Environments by Trained Agent Exploration

Naser Kazemi, Nedko Savov, Danda Pani Paudel, Luc Van Gool, [Webpage](#), [arXiv](#)

NeurIPS Workshop on D3S3

Sep. 2024

Relevant Courses

- Reinforcement Learning
- Convex Optimization
- Information Theory
- High Dimensional Probability (N/A)
- Mathematical Analysis 1
- Algebra 1
- Game Theory
- Design of Algorithms
- Combinatorics (N/A)

Awards and Honors

July. 2020 National University Entrance Exam: Ranked 118th

Feb. 2018 European Mathematical Cup: Gold Medal

Github Projects

- **GenieRedux**, Pytorch Implementation of Genie Model, [Link](#)
- **RL Practical and Theoretical Assignments**, Reinforcement Learning Course, PyTorch, [Link](#)
- **Convex Optimization Algorithms**, Convex Optimization Course, CVXPY, [Link](#)
- **Massive Data Algorithms**, Massive Data Analysis Course, PySpark, [Link](#)
- **Rope Motion Simulation**, Digital Systems Design, Verilog & C++, [Link](#)
- **C-Minus Compiler**, Compiler Design Course, Python, [Link](#)
- **Yu-Gi-Oh! Trading Card Game**, Advanced Programming Project, Java, [Link](#)

Side Courses

- **Functional Programming Principles in Scala**, Coursera, by EPFL
- **Functional Program Design in Scala**, Coursera, by EPFL

Technical Skills

Programming	Java, C/C++, C# , Python, Scala, CUDA, R
Libraries & Frameworks	PyTorch, Scikit-Learn, TensorFlow, Jax, CVXPY, DEAP, Hadoop, Spark, Kafka, InfluxDB, Django
Softwares	Linux, Xilinx, L ^A T _E X, Git, Docker
Languages	Persian(Native), English (IELTS Band 8)

Voluntary Services

Student Scientific Chapter

Computer Engineering Department, Sharif University of Technology
Central Member

Tehran

July. 2022 - Sep. 2023

Rayanesh Student Journal

Computer Engineering Department, Sharif University of Technology
Writer

Tehran

2021 - Now

ICPC 2023

Sharif University of Technology
Executive Staff

Tehran

June 2024

Teaching Assistance (Unpaid)

Sharif University of Technology

- **Linear Algebra** | Head Teaching Assistant, Managing a team of 20+ TAs
- **Game Theory** | Head Teaching Assistant, Managing a team of 10+ TAs
- **Probability and Statistics** | Workshop
- **Data Structures and Algorithms** | Workshop
- **Convex Optimization**
- **Machine Learning**